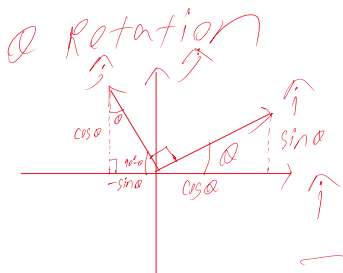
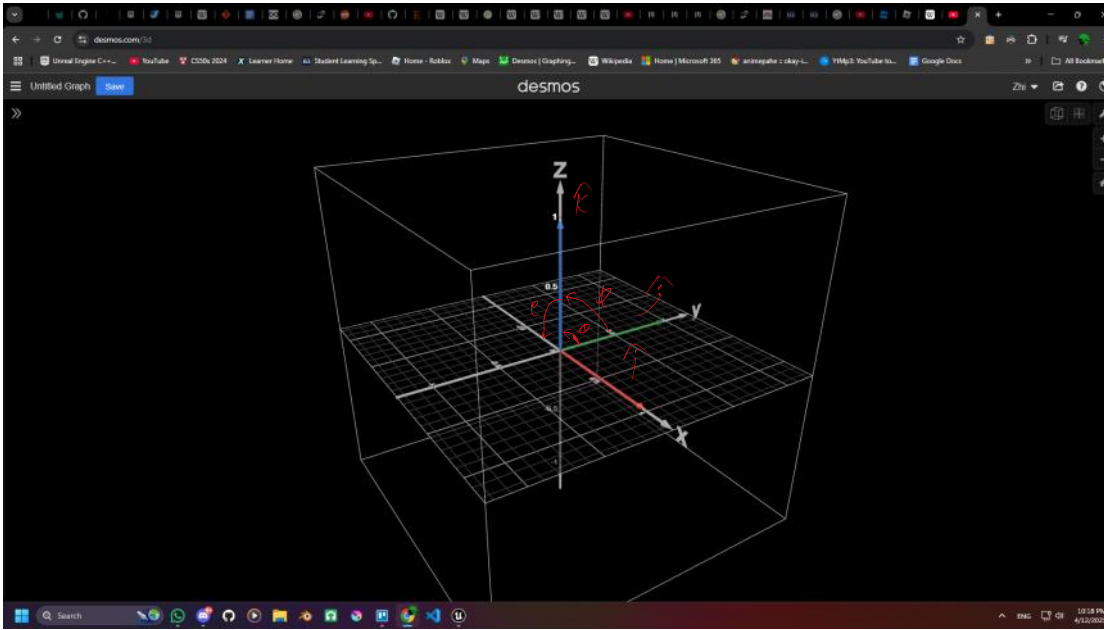


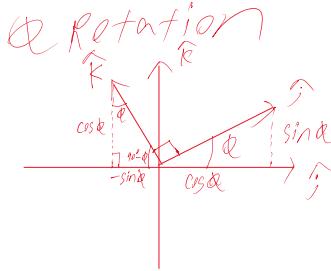
Rotation matrix (Rework) (Final)

Sunday, April 13, 2025 7:50 PM

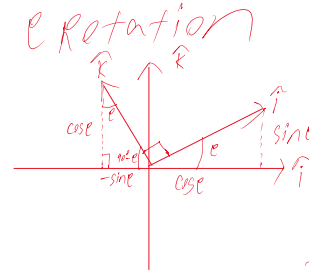
Let the Yaw, Pitch and Roll rotation angles be θ , ϕ and ρ respectively



$$\begin{bmatrix} \cos \rho & -\sin \rho & 0 \\ \sin \rho & \cos \rho & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix}$$



$$\begin{bmatrix} \cos \phi & 0 & -\sin \phi \\ 0 & 1 & 0 \\ \sin \phi & 0 & \cos \phi \end{bmatrix}$$

$$\begin{bmatrix} \cos \rho & -\sin \rho & 0 \\ \sin \rho & \cos \rho & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix}$$

$$\begin{bmatrix} \cos \phi & 0 & -\sin \phi \\ 0 & 1 & 0 \\ \sin \phi & 0 & \cos \phi \end{bmatrix}$$

$$= \begin{bmatrix} \cos \rho & -\sin \rho & 0 \\ \sin \rho & \cos \rho & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos \phi & 0 & -\sin \phi \\ -\sin \phi \sin \rho & \cos \phi & -\sin \phi \cos \rho \\ \cos \phi \sin \rho & \sin \phi & \cos \phi \cos \rho \end{bmatrix}$$

$$= \cos \rho \cos \phi$$

$$\begin{matrix}
 \left[\begin{matrix} 0 & 0 & 1 \end{matrix} \right] \left[\begin{matrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{matrix} \right] \\
 = \left[\begin{matrix} \sin \theta \sin \phi \sin \rho + \cos \theta \cos \rho & -\sin \theta \cos \phi & \sin \theta \sin \phi \cos \rho - \cos \theta \sin \rho \\ \sin \theta \cos \rho - \cos \theta \sin \phi \sin \rho & \cos \theta \cos \phi & -\cos \theta \sin \phi \cos \rho - \sin \theta \sin \rho \\ \cos \phi \sin \rho & \sin \phi & \cos \phi \cos \rho \end{matrix} \right]
 \end{matrix}$$

$$\begin{bmatrix} \sin \theta \sin \phi \sin \rho + \cos \theta \cos \rho & -\sin \theta \cos \phi & \sin \theta \sin \phi \cos \rho - \cos \theta \sin \rho \\ \sin \theta \cos \rho - \cos \theta \sin \phi \sin \rho & \cos \theta \cos \phi & -\cos \theta \sin \phi \cos \rho - \sin \theta \sin \rho \\ \cos \phi \sin \rho & \sin \phi & \cos \phi \cos \rho \end{bmatrix}$$

```

\begin{bmatrix}
\sin\theta\sin\phi\sin\rho+\cos\theta\cos\rho
&\sin\theta\cos\phi
&\sin\theta\sin\phi\cos\rho-\cos\theta\sin\rho\\
\sin\theta\cos\rho-\cos\theta\sin\phi\sin\rho
&\cos\theta\cos\phi
&-\cos\theta\sin\phi\cos\rho-\sin\theta\sin\rho\\
\cos\phi\sin\rho
&\sin\phi
&\cos\phi\cos\rho
\end{bmatrix}

```

LES GO IT fina works
<https://www.desmos.com/3d/rzynh2dgrj?invertedColors=true>